

08 — The Register (the book of record)

Audience: every backend engineer. This is the service you will spend most of your time in. **Companion docs:** [15 Data model](#) · [04 Running flows](#) · [07 RBAC](#) **Code:** `services/register/`

1. What the Register is, and what it refuses to be

The Register is **the only service that owns business data**. Every other service in PRISM is stateless and reaches the data through this API. That single fact buys three things:

- A bug in the dashboard, the news radar, VocX or a workflow **cannot corrupt the book**.
- There is exactly one place where tenancy, concurrency, audit and lifecycle rules are enforced — so they cannot drift between services.
- Backup and restore have one subject.

What it deliberately is *not*: it holds no business logic that belongs to a process. A multi-step change with a human decision point lives in a Temporal workflow ([05](#)), and the Register provides the transactional primitives that workflow calls.

2. Layout

```
services/register/app/
├─ api/           34 routers – the REST surface
├─ authz/        the engine that applies RBAC + policy to a request
├─ core/         config, security/RequestContext, pagination, router, people, reconciliation
├─ db/           session, RLS application
├─ models/       20 modules → ~43 tables
├─ repositories/ CRUDRepository + per-domain repositories
├─ schemas/      Pydantic create/update/read per resource
├─ seed/         reference data, atlas_data.json, MIS xlsx importer, ledger round-trip
└─ storage/      MinIO / object storage
```

3. The generic CRUD router — read this before adding a resource

`app/api/crud_router.py` builds an identical, well-tested surface for every table:

```
POST  /           create (idempotent via Idempotency-Key)
GET   /           list (search, keyset pagination, whitelisted filters)
GET   /{id}       read one
PATCH /{id}      partial update (optimistic concurrency)
DELETE /{id}      soft delete (optimistic concurrency)
POST  /{id}/restore undo a soft delete
```

"Doing this once means the concurrency, tenancy, auditing and error semantics are identical and correct across all 18 tables rather than copy-pasted 18 times."

A resource is declared as a `ResourceSpec`, not written by hand:

```
ResourceSpec(  
  name="entity", prefix="/v1/entities", tags=["Entities"],  
  repo=CRUDRepository(Entity,  
    searchable=["legal_name", "code", "display_name", "cin"],  
    filterable=["sector", "lens", "register_status", "entity_type",  
               "state", "promoter_group_code", "code"]),  
  create_schema=s.EntityCreate, update_schema=s.EntityUpdate, read_schema=s.EntityRead,  
  filterable=[...],  
  subject_type="Entity",      # what the policy engine keys lifecycle rules on  
  view_name="clients",        # which view-matrix row governs read access  
  pre_delete=_entity_pre_delete,  
)
```

Spec field	What it controls
<code>subject_type</code>	The key into <code>STAGE_VOCAB</code> / <code>MANDATORY_FOR_STAGE</code> / <code>FIELD_LOCKS</code> . Must match the API subject string — a mismatch silently disables every lifecycle rule for that resource.
<code>view_name</code>	Which row of the view matrix governs read access
<code>write_operation</code>	Which operation governs writes (defaults per resource)
<code>company_scoped</code> , <code>parent_scope</code>	How scoping is derived for SCOPED users
<code>pre_write</code> , <code>pre_delete</code>	Domain hooks (e.g. one-person-one-mailbox on <code>Person</code>)
<code>include_create/u</code> <code>pdate/delete</code>	Turn off a verb for a read-only projection

The `subject_type` warning is not theoretical. From `policy.py`: *"A key that does not match the API subject silently disables every mandatory/lock rule for that resource, so these MUST stay in lock-step with the resource registry."* Integration tests PATCH the real `/v1/lending` and `/v1/syndication` routes precisely to guard this.

Resources built from the factory

`/v1/entities` · `/v1/people` · `/v1/counterparties` · `/v1/leads` · `/v1/deals` · `/v1/lending` ·
`/v1/syndication` · `/v1/syndication-lenders` · `/v1/asset-monetisation` · `/v1/financials` ·
`/v1/contracts-assets` · `/v1/interactions` · `/v1/external-intelligence` · `/v1/monitoring` ·
`/v1/documents` · `/v1/document-checklist`

4. The custom routers

Anything that is not plain CRUD gets its own module in `app/api/`.

Router	Surface	Why it is not CRUD
<code>rbac.py</code>	<code>/v1/assignments</code> · <code>/v1/requests</code> · <code>/v1/requests/{id}/approve</code> · <code>/v1/leads/{id}/convert</code> · <code>/v1/authz/check</code>	Assignment and approval are authority decisions; conversion is transactional
<code>reconciliation.py</code>	<code>/v1/reconciliation</code>	Import quarantine queue — see §8
<code>imports.py</code>	<code>/v1/import/atlas-xlsx</code>	Spreadsheet ingest
<code>export.py</code> , <code>export_ledger.py</code>	<code>/v1/export/{json,excel,counts}</code> · <code>/v1/export/ledger-xlsx</code>	Scoped bulk read
<code>custom.py</code>	<code>/v1/audit</code> · <code>/v1/entities/{id}/dossier</code> · <code>/v1/documents/upload</code> · <code>/v1/ref</code>	Composed reads and binary upload
<code>closure.py</code>	<code>/v1/deals/{id}/close</code> · <code>/v1/deals/{id}/open-items</code>	Closure has preconditions
<code>lms.py</code>	<code>/v1/lending/{id}/loan-account</code> (+ entries, conditions, accrue, interest-preview)	Loan servicing ledger
<code>tranches.py</code>	<code>/v1/lending/{id}/tranches</code> · <code>/v1/bookings/pending</code>	Drawdown booking
<code>sanction.py</code>	<code>/v1/internal/cam-reports...</code>	The CAM workbench
<code>covenants.py</code>	<code>/v1/covenants</code> · <code>/v1/monitoring/{id}/result</code> · <code>/waive</code>	Covenant lifecycle
<code>ews.py</code>	<code>/v1/ews-cases</code> (+ assign, escalate, close, note)	Early-warning cases
<code>cpcs.py</code>	<code>/v1/internal/cpcs-checklists...</code>	CP/CS maker-checker
<code>handover.py</code>	<code>/v1/internal/handover-packages...</code>	Advaya handover, immutable package
<code>advaya.py</code>	<code>/v1/internal/advaya-handoffs</code>	The boundary record
<code>decisions.py</code>	<code>/v1/internal/decisions...</code>	Durable single-winner workflow decisions
<code>notifications.py</code> , <code>notify.py</code>	<code>/v1/notifications</code> · <code>/v1/internal/notifications/deliveries...</code>	Outbox + delivery claim
<code>documents_lifecycle.py</code>	<code>/v1/documents/{id}/{validate,reject,replace}</code> · expiry sweep	Document states
<code>evidence.py</code>	<code>/v1/evidence</code>	Governance evidence with break-glass
<code>calendar.py</code>	<code>/v1/calendar-events</code> (+ complete, cancel)	Scheduling
<code>followups.py</code>	<code>/v1/internal/follow-ups</code>	Derived work items
<code>series.py</code>	<code>/v1/internal/number-series</code>	Tracker numbering
<code>people_sync.py</code>	<code>/v1/internal/people/{sync-access,handover}</code>	Directory ↔ Access reconciliation
<code>tenants.py</code>	<code>/v1/tenants</code>	Tenant administration (admin credential on every method, reads

Router	Surface	Why it is not CRUD
		included)
<code>entity_rules.py</code> , <code>people_rules.py</code>	—	Pre-write/pre-delete domain hooks

The `/v1/internal/` convention

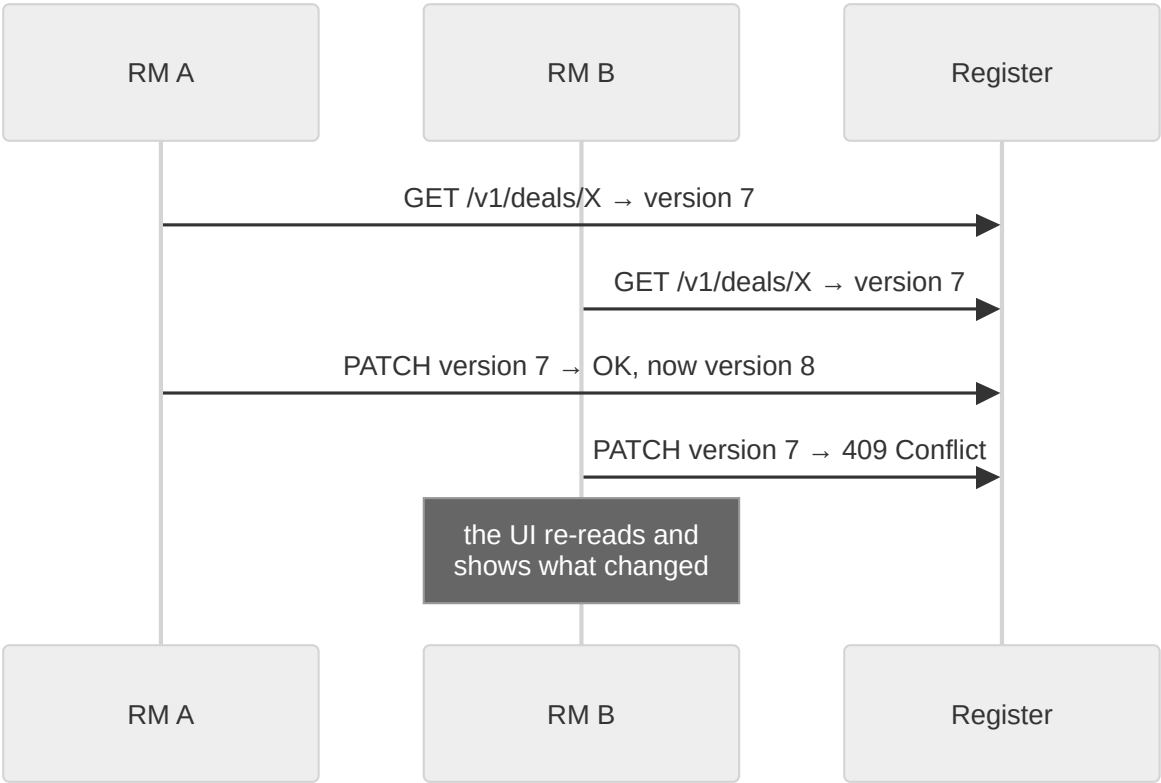
Routes under `/v1/internal/` are the **machine lane**: they enforce *named service principals* rather than human RBAC. They are reachable from outside only through the edge's dedicated location:

```
location /machine/v1/internal/ {
    resolver 127.0.0.11 valid=10s;
    rewrite ^/machine(/v1/internal/.*)$ $1 break;
    proxy_pass $register_upstream;
}
```

So Advaya's boundary calls, cron sweeps and the notifier reach the internal API through the same one door — nothing needs a direct service port. Human CRUD stays gateway-only.

5. Concurrency: optimistic, and visible

Every mutable row carries a `version`. `PATCH` and `DELETE` require the caller's version to match; a mismatch is a **409**, not a silent overwrite.



Two RMs editing the same deal produce a visible conflict, never a lost update. See `docs/adr/0003-optimistic-concurrency.md`.

6. Soft delete

`DELETE` is a soft delete; `POST /{id}/restore` undoes it. `delete_row` is an **Admin-only** operation in the matrix, and the gateway gates it at the door:

```
("DELETE", r"^/v1/(?!users|assignments|requests)[^/]+/[^/]+$", "delete_row"),
```

Restores and deletes are among the operations that revalidate against Access *online* in production posture — a revoked Admin cannot delete on a stale signed context.

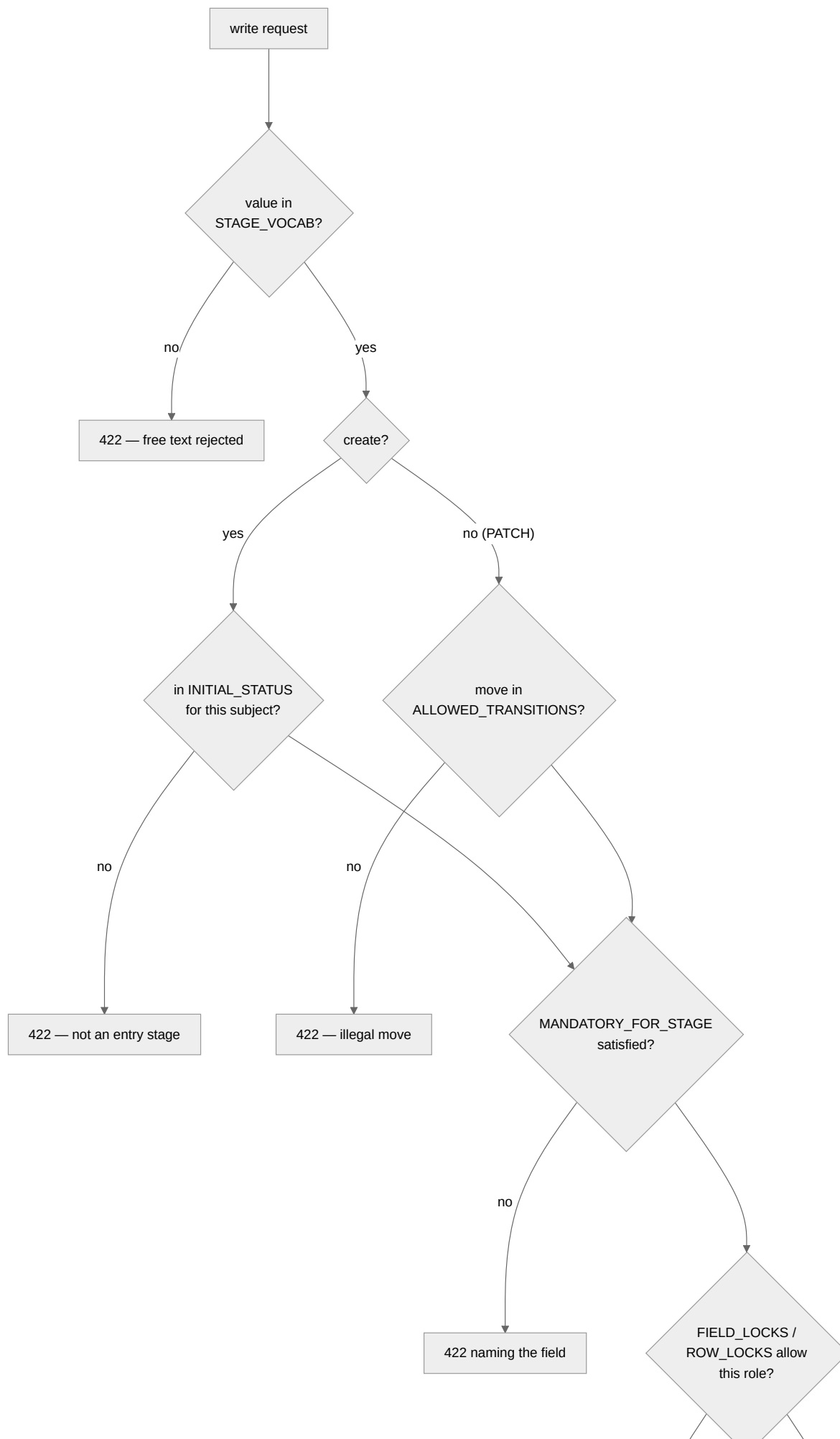
7. Idempotency

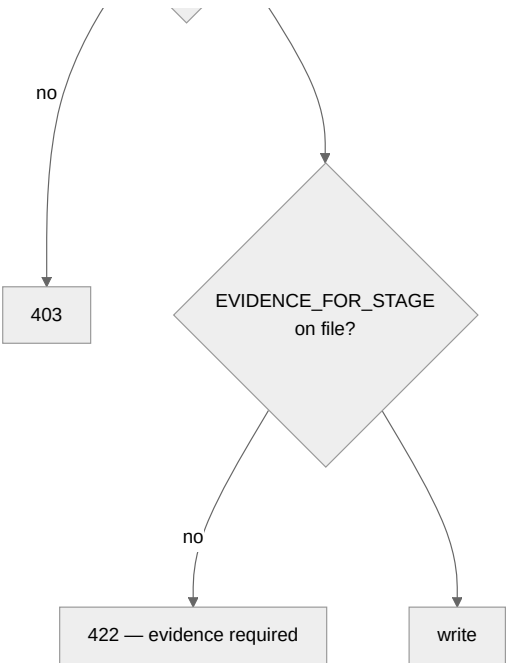
`POST` accepts `Idempotency-Key`. The key and a hash of the payload are recorded in `idempotency_keys`; a replay returns the original result rather than creating a second row. This is what makes Temporal's unlimited-retry policy safe.

Known gap: `POST /v1/{subject}/{id}/interactions` does not take an idempotency key. A retried interaction write can duplicate a timeline row.

8. Lifecycle enforcement

The Register is where the lifecycle rules from `packages/evam-backend-core/evam_backend_core/lifecycle.py` and `policy.py` actually bite.





The vocabularies

Subject	Field	Values
Lead	status	Active · On Hold · Dropped · Converted
Deal	stage	New Inquiry · In Screening · In Pipeline · On Hold · Screened Out · Closed Won · Closed Lost · Dropped
Lending	stage	Data Awaited · Diligence · Note Circulated · Sanctioned · CP/CS Completed · Ready for Disbursement · Disbursed · Rejected · On Hold
Syndication	status	Deal Sourced · Docs Pending · IM in Prep · IM Circulated · Queries Received · IP Received · Sanctioned · Disbursed · On Hold · Withdrawn · Rejected · Dropped
AssetMonetisation	status	Teaser Prepared · Teaser Shared · In Discussion · NBO Received · BO Received · SPA / Documentation · Closed · Dropped

Syndication has no **Diligence**. Lending does. The two vocabularies are separate on purpose and mapping a spreadsheet across them is a common source of import errors.

Entry states

Subject	May be born at
Lead	Active · On Hold · Dropped
Deal	New Inquiry · In Screening · In Pipeline · On Hold
Lending	Data Awaited · Diligence
Syndication	Deal Sourced · Docs Pending · IM in Prep
AssetMonetisation	Teaser Prepared · Teaser Shared · In Discussion

Every later stage — including working states like **Note Circulated** or **IM Circulated**, and every terminal — must be *stepped* to, or arrive through a separately-audited import.

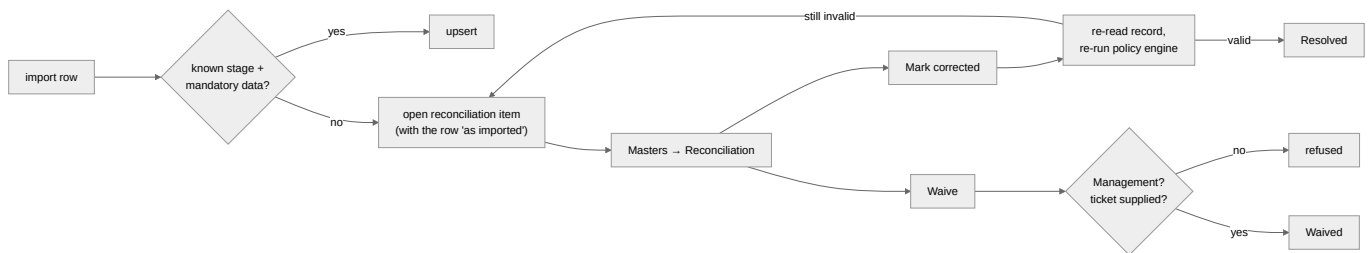
Reference vocabularies are data, not code

`app/seed/refdata.py` seeds `ref_values`, served from `GET /v1/ref`. Front-ends fetch the dropdowns rather than shipping their own copy, **so a vocabulary change is a data change, not a browser redeploy**. Person names are never seeded — `/v1/ref` merges the live people directory in.

A register-side test cross-checks `refdata.py` against `lifecycle.py` so the dropdown and the enforcement cannot drift.

9. Import reconciliation

An import must never write a row it knows to be wrong, and must never silently drop it. So a row whose stage is unknown, or which lacks that stage's mandatory data, becomes an `import_reconciliation_items` row.



The authority split is deliberate and enforced in `api/reconciliation.py`:

- **Working the queue** — Admin or Management.
- **Waiving** — Management **only**, and a **ticket reference is required**.

An Admin may close a corrected item but may not decide that a record stays incomplete. And "corrected" is not a status flip: resolution re-reads the record and re-runs the policy engine, so it cannot be claimed for a row that is still wrong.

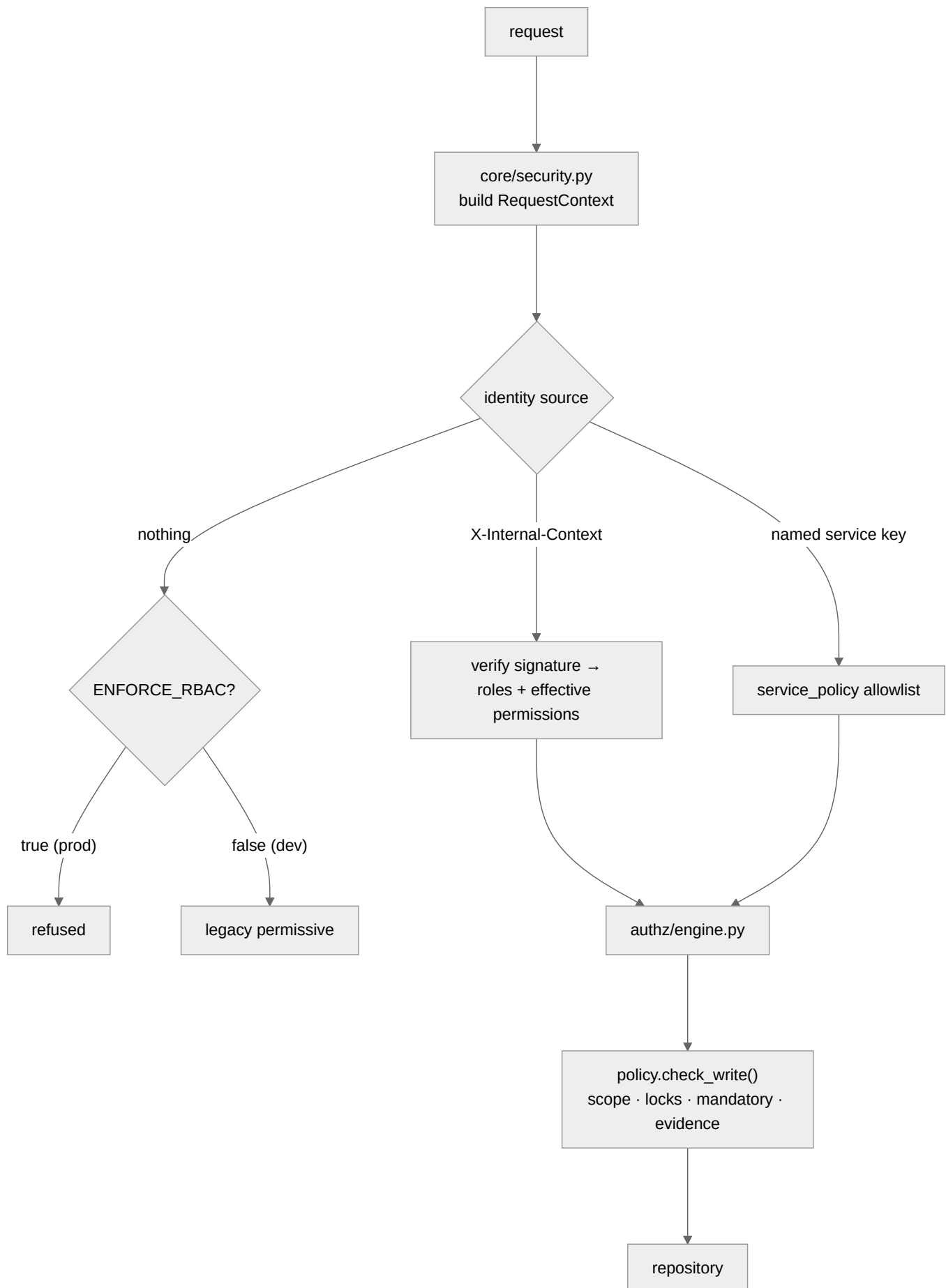
10. Tenancy and RLS

Every row carries `tenant_id`. Two independent mechanisms enforce isolation:

1. Application scoping from `X-Tenant` or the signed context.
2. **PostgreSQL row-level security**, applied and force-converged at startup when `REGISTER_ENFORCE_RLS=true` (`app/db/apply_qls.py`).

RLS is **fail-closed**. If the policy cannot be established the service refuses to serve. The point is that a forgotten `WHERE` clause in application code is contained by the database rather than becoming a cross-tenant leak.

11. Authorization inside the Register



The Register **never re-derives** authorization from a static copy of the matrix — it enforces the effective grant carried in the verified signed context, which is what Access resolved at that moment.

For sensitive operations with `REGISTER_ONLINE_REVALIDATION=true` (delete/restore, assignments, governed imports, evidence break-glass) it additionally calls Access **online** and fails closed (503) if Access is down. A revocation therefore takes effect immediately for the actions that cannot be undone, regardless of any signed context still within its TTL.

12. Pagination, search and filtering

- **Keyset pagination**, not offset — stable under concurrent writes.
- **Search** is restricted to the `searchable` columns declared on the repository.
- **Filters are whitelisted** per resource. An unknown filter is not ignored — it is **rejected**. Silently dropping an unrecognised filter would widen a result set a caller believed was narrowed, which is a data-leak shape.

13. Seeding and data loading

Entry point	Purpose
<code>app/seed/refdata.py</code>	Reference vocabularies → <code>ref_values</code>
<code>app/seed/loader.py</code>	<code>atlas_data.json</code> demo/prototype data
<code>app/seed/from_xlsx.py</code>	The MIS consolidated workbook → the whole book
<code>app/seed/ledger_xlsx.py</code>	The desk's Excel in <i>and</i> out, round-trip

```
# Replace everything from a workbook
python -m app.seed.from_xlsx data/Evam_ATLAS_MIS_Consolidated_v4.xlsx

# Merge / upsert instead of replacing
python -m app.seed.from_xlsx <path> --no-truncate
```

Sheets consumed: **Deals** (with the three product flags), **Lending Tracker**, **Syndication** (one tracker + per-bank lender rows), **Asset Mon** (one row per *mandate* — a company may have several), **Mandate Tracker**. Every distinct Company Name across all sheets becomes one entity; distinct RMs/analysts become people; distinct banks become counterparties.

A name typo creates a second company. The importer is entity-centric and matches by name. This is the single most common import defect.

`deal_id` on a tracker row comes from `deal_by_entity.get(entity)` — `None` when a company appears on a tracker sheet but not on the Deals sheet. Such rows are legal (`deal_id` is nullable) and display correctly; they are mandates with no deal record.

14. Adding a resource — the checklist

- ☐ Model in `app/models/`, with `tenant_id`, `version`, timestamps and soft-delete columns.
- ☐ Migration.
- ☐ Pydantic `Create` / `Update` / `Read` in `app/schemas/resources.py`.
- ☐ `ResourceSpec` in `app/api/resources.py` — set `subject_type` and `view_name` correctly.
- ☐ If it has a lifecycle: add to `STAGE_VOCAB`, `INITIAL_STATUS`, `ALLOWED_TRANSITIONS`, and `_STAGE_FIELD` in `policy.py`.
- ☐ If it has stage-mandatory fields or locks: `MANDATORY_FOR_STAGE`, `FIELD_LOCKS`, `ROW_LOCKS`.
- ☐ Reference vocabulary → `app/seed/refdata.py` (and the cross-check test will hold you to it).
- ☐ Gateway `routes_map.py` entry if the route should be gated at the door.
- ☐ Machine access → `service_policy.py` if a service writes it.
- ☐ RLS policy in `app/db/apply_qls.py`.
- ☐ UI service module in `services/atlas/ui/src/services/`.
- ☐ Tests: CRUD, concurrency (no lost updates), scope, and the lifecycle refusals.